

Отчёт о лабораторной работе

Лабораторная работа 6

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1 Цель работы

Развить навыки администрирования ОС Linux. Получить первое практическое знакомство с технологией SELinux. Проверить работу SELinux на практике совместно с веб-сервером Apache.

2 Выполнение лабораторной работы

Проверим, что SELinux работает в режиме enforcing (рис. 2.1).

```
[nsandriushin@nsandriushin ~]$ getenforce
Enforcing
[nsandriushin@nsandriushin ~]$ sestatus
SELinux status:                enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:              targeted
Current mode:                   enforcing
Mode from config file:          enforcing
Policy MLS status:              enabled
Policy deny_unknown status:     allowed
Memory protection checking:     actual (secure)
Max kernel policy version:      33
[nsandriushin@nsandriushin ~]$ service httpd status
Redirecting to /bin/systemctl status httpd.service
o httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: inactive (dead)
   Docs: man:httpd.service(8)
[nsandriushin@nsandriushin ~]$ service httpd start
Redirecting to /bin/systemctl start httpd.service
[nsandriushin@nsandriushin ~]$ ps auxZ | grep httpd
system_u:system_r:httpd_t:s0    root      42803  0.3  0.2  21032 11372 ?        Ss   21:38   0:00 /usr/sb
in/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0    apache    42813  0.0  0.1  22908  7384 ?        S    21:38   0:00 /usr/sb
in/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0    apache    42815  1.0  0.4  2227512 17172 ?       Sl   21:38   0:00 /usr/sb
in/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0    apache    42816  0.6  0.3  2358648 15164 ?       Sl   21:38   0:00 /usr/sb
in/httpd -DFOREGROUND
system_u:system_r:httpd_t:s0    apache    42817  0.8  0.4  2227512 17180 ?       Sl   21:38   0:00 /usr/sb
in/httpd -DFOREGROUND
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 nsandri+ 43017 0.0  0.0  221664 2432 pts/0 S+   21:38
0:00 grep --color=auto httpd
```

Рис. 2.1: Режим SELinux

Посмотрим переключатели SELinux для httpd (рис. 2.2).

```

[nsandriushin@nsandriushin ~]$ sestatus -b | grep httpd
httpd_anon_write                off
httpd_builtin_scripting         on
httpd_can_check_spam            off
httpd_can_connect_ftp           off
httpd_can_connect_ldap          off
httpd_can_connect_mythtv        off
httpd_can_connect_zabbix        off
httpd_can_manage_courier_spool   off
httpd_can_network_connect       off
httpd_can_network_connect_cobbler off
httpd_can_network_connect_db    off
httpd_can_network_memcache      off
httpd_can_network_relay         off
httpd_can_sendmail              off
httpd_dbus_avahi                off
httpd_dbus_sssd                 off
httpd_dontaudit_search_dirs     off
httpd_enable_cgi                on
httpd_enable_ftp_server         off
httpd_enable_homedirs           off
httpd_execmem                   off
httpd_graceful_shutdown         off
httpd_manage_ipa                off
httpd_mod_auth_ntlm_winbind     off
httpd_mod_auth_pam              off
httpd_read_user_content         off
httpd_run_ipa                   off
httpd_run_preupgrade            off
httpd_run_stickshift            off
httpd_serve_cobbler_files       off
httpd_setrlimit                 off
httpd_ssi_exec                  off
httpd_sys_script_anon_write     off
httpd_tmp_exec                  off
httpd_tty_comm                  off
httpd_unified                   off
httpd_use_cifs                  off
httpd_use_fusefs                off
httpd_use_gpg                   off
httpd_use_nfs                   off
httpd_use_opencryptoki          off
httpd_use_openstack             off
httpd_use_sasl                  off
httpd_verify_dns                off

```

Рис. 2.2: Переключатели SELinux

Посмотрим на количество типов, пользователей и ролей с помощью команды `seinfo` (рис. 2.3).

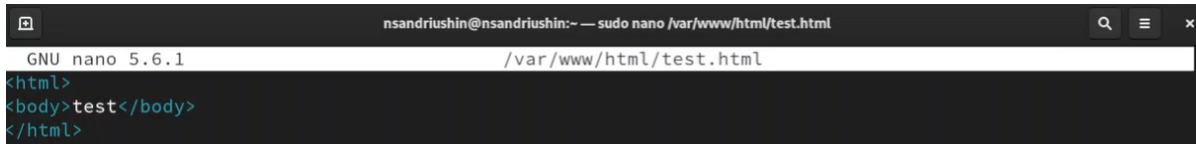
```

[nsandriushin@nsandriushin ~]$ seinfo
Statistics for policy file: /sys/fs/selinux/policy
Policy Version:          33 (MLS enabled)
Target Policy:           selinux
Handle unknown classes:  allow
Classes:                 135
Sensitivities:           1
Types:                   5169
Users:                   8
Booleans:                358
Allow:                   65633
Auditallow:              176
Type_trans:              271851
Type_member:             37
Role_allow:              40
Constraints:             70
MLS Constrain:           72
Permissives:             1
Defaults:                7
Allowxperm:              0
Auditallowxperm:         0
Ibendportcon:            0
Initial SIDs:            27
Genfscon:                109
Netifcon:                0
Permissions:             457
Categories:              1024
Attributes:              259
Roles:                   15
Cond. Expr.:             390
Neverallow:              0
Dontaudit:               8703
Type_change:             94
Range_trans:             5931
Role_trans:              417
Validatetrans:           0
MLS Val. Tran:           0
Polcap:                  6
Typebounds:              0
Neverallowxperm:         0
Dontauditxperm:          0
Ibpkeycon:               0
Fs_use:                  35
Portcon:                 665
Nodecon:                 0
[nsandriushin@nsandriushin ~]$ ls -lZ /var/www
total 0
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_script_exec_t:s0 6 Jan 22 03:25 cgi-bin
drwxr-xr-x. 2 root root system_u:object_r:httpd_sys_content_t:s0      6 Jan 22 03:25 html
[nsandriushin@nsandriushin ~]$ ls -lZ /var/www/html
total 0
[nsandriushin@nsandriushin ~]$ sudo nano /var/www/html/test.html

```

Рис. 2.3: seinfo

Создадим файл /var/www/html/test.html и заполним его одним словом (рис. 2.4).



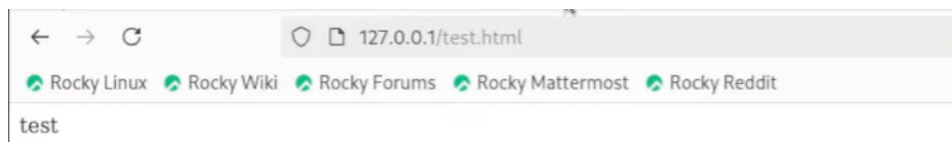
```

nsandriushin@nsandriushin:~ -- sudo nano /var/www/html/test.html
GNU nano 5.6.1 /var/www/html/test.html
<html>
<body>test</body>
</html>

```

Рис. 2.4: /var/www/html/test.html

При открытии браузера открывается созданная нами страница (рис. 2.5).



```

127.0.0.1/test.html
test

```

Рис. 2.5: Открытие страницы

Далее посмотрим на метки в директории /var/www/html/ и попробуем сменить метку созданного нами файла и посмотрим логи (рис. 2.6).


```

[nsandriushin@nsandriushin ~]$ ls -lZ /var/www/html
total 4
-rw-r--r--. 1 root root unconfined_u:object_r:httpd_sys_content_t:s0 33 May  3 21:43 test.html
[nsandriushin@nsandriushin ~]$ man httpd_selinux
No manual entry for httpd_selinux
[nsandriushin@nsandriushin ~]$ man httpd selinux
--Man-- next: selinux(8) [ view (return) | skip (Ctrl-D) | quit (Ctrl-C) ]
^C
[nsandriushin@nsandriushin ~]$ sudo chcon -t samba_share_t /var/www/html/test.html
[sudo] password for nsandriushin:
[nsandriushin@nsandriushin ~]$ ls -Z /var/www/html/test.html
unconfined_u:object_r:samba_share_t:s0 /var/www/html/test.html
[nsandriushin@nsandriushin ~]$ ls -l /var/www/html/test.html
-rw-r--r--. 1 root root 33 May  3 21:43 /var/www/html/test.html
[nsandriushin@nsandriushin ~]$ tail /var/log/messages
tail: cannot open '/var/log/messages' for reading: Permission denied
[nsandriushin@nsandriushin ~]$ sudo tail /var/log/messages
May  3 21:49:45 nsandriushin systemd[1]: Created slice Slice /system/dbus-:1.1-org.fedoraproject.Setrouble
shootPrivileged.
May  3 21:49:45 nsandriushin systemd[1]: Started dbus-:1.1-org.fedoraproject.SetroubleshootPrivileged@0.se
rvice.
May  3 21:49:47 nsandriushin setroubleshoot[43770]: SELinux is preventing /usr/sbin/httpd from getattr acc
ess on the file /var/www/html/test.html. For complete SELinux messages run: sealert -l e9059f36-4ec2-44a6-
8a1b-6f89c8870b3d
May  3 21:49:47 nsandriushin setroubleshoot[43770]: SELinux is preventing /usr/sbin/httpd from getattr acc
ess on the file /var/www/html/test.html.#012#012***** Plugin restorecon (92.2 confidence) suggests ****
*****#012#012If you want to fix the label. #012/var/www/html/test.html default label should
be httpd_sys_content_t.#012Then you can run restorecon. The access attempt may have been stopped due to i
nsufficient permissions to access a parent directory in which case try to change the following command acc
ordingly.#012Do#012# /sbin/restorecon -v /var/www/html/test.html#012#012***** Plugin public_content (7.83
confidence) suggests *****#012#012If you want to treat test.html as public content#012Th
en you need to change the label on test.html to public_content_t or public_content_rw_t.#012Do#012# seman
age fcontext -a -t public_content_t '/var/www/html/test.html'#012# restorecon -v '/var/www/html/test.html'#
012#012***** Plugin catchall (1.41 confidence) suggests *****#012#012If you believ
e that httpd should be allowed getattr access on the test.html file by default.#012Then you should report
this as a bug.#012You can generate a local policy module to allow this access.#012Do#012allow this access
for now by executing:#012# ausearch -c 'httpd' --raw | audit2allow -M my-httpd#012# semodule -X 300 -i my-
httpd.pp#012
May  3 21:49:47 nsandriushin setroubleshoot[43770]: SELinux is preventing /usr/sbin/httpd from getattr acc
ess on the file /var/www/html/test.html. For complete SELinux messages run: sealert -l e9059f36-4ec2-44a6-
8a1b-6f89c8870b3d
May  3 21:49:47 nsandriushin setroubleshoot[43770]: SELinux is preventing /usr/sbin/httpd from getattr acc
ess on the file /var/www/html/test.html.#012#012***** Plugin restorecon (92.2 confidence) suggests ****
*****#012#012If you want to fix the label. #012/var/www/html/test.html default label should
be httpd_sys_content_t.#012Then you can run restorecon. The access attempt may have been stopped due to i
nsufficient permissions to access a parent directory in which case try to change the following command acc
ordingly.#012Do#012# /sbin/restorecon -v /var/www/html/test.html#012#012***** Plugin public_content (7.83
confidence) suggests *****#012#012If you want to treat test.html as public content#012Th
en you need to change the label on test.html to public_content_t or public_content_rw_t.#012Do#012# seman
age fcontext -a -t public_content_t '/var/www/html/test.html'#012# restorecon -v '/var/www/html/test.html'#
012#012***** Plugin catchall (1.41 confidence) suggests *****#012#012If you believ
e that httpd should be allowed getattr access on the test.html file by default.#012Then you should report
this as a bug.#012You can generate a local policy module to allow this access.#012Do#012allow this access
for now by executing:#012# ausearch -c 'httpd' --raw | audit2allow -M my-httpd#012# semodule -X 300 -i my-

```

Рис. 2.6: Смена метки

После смены метки доступ к странице был запрещён (рис. 2.7).

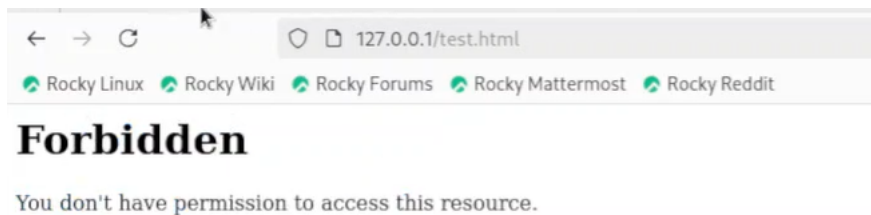


Рис. 2.7: Доступ запрещён

Поменяем порт на 81 вместо 80 (рис. 2.8).

```
GNU nano 5.6.1 /etc/httpd/conf/httpd.conf Modified
# This is the main Apache HTTP server configuration file. It contains the
# configuration directives that give the server its instructions.
# See <URL:http://httpd.apache.org/docs/2.4/> for detailed information.
# In particular, see
# <URL:http://httpd.apache.org/docs/2.4/mod/directives.html>
# for a discussion of each configuration directive.
#
# See the httpd.conf(5) man page for more information on this configuration,
# and httpd.service(8) on using and configuring the httpd service.
#
# Do NOT simply read the instructions in here without understanding
# what they do. They're here only as hints or reminders. If you are unsure
# consult the online docs. You have been warned.
#
# Configuration and logfile names: If the filenames you specify for many
# of the server's control files begin with "/" (or "drive:/" for Win32), the
# server will use that explicit path. If the filenames do *not* begin
# with "/", the value of ServerRoot is prepended -- so 'log/access_log'
# with ServerRoot set to '/www' will be interpreted by the
# server as '/www/log/access_log', where as '/log/access_log' will be
# interpreted as '/log/access_log'.
#
# ServerRoot: The top of the directory tree under which the server's
# configuration, error, and log files are kept.
#
# Do not add a slash at the end of the directory path. If you point
# ServerRoot at a non-local disk, be sure to specify a local disk on the
# Mutex directive, if file-based mutexes are used. If you wish to share the
# same ServerRoot for multiple httpd daemons, you will need to change at
# least PidFile.
#
ServerRoot "/etc/httpd"
#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts. See the httpd.service(8) man
# page for more information.
#Listen 12.34.56.78:80
Listen 81
```

Рис. 2.8: Смена порта

Перезагрузим службу. После этого мы не сможем зайти на нашу страницу. Об этом также будут писать логи (рис. 2.9).

```
[nsandriushin@nsandriushin ~]$ sudo nano /etc/httpd/conf/httpd.conf
[nsandriushin@nsandriushin ~]$ systemctl restart httpd
[nsandriushin@nsandriushin ~]$ service httpd restart
Redirecting to /bin/systemctl restart httpd.service
[nsandriushin@nsandriushin ~]$ tail -nl /var/log/messages
tail: invalid number of lines: 'l'
[nsandriushin@nsandriushin ~]$ sudo tail -nl /var/log/messages
tail: invalid number of lines: 'l'
[nsandriushin@nsandriushin ~]$ sudo tail -n /var/log/messages
tail: invalid number of lines: '/var/log/messages'
[nsandriushin@nsandriushin ~]$ sudo tail /var/log/messages
May  3 21:53:10 nsandriushin systemd[1]: Started The Apache HTTP Server.
May  3 21:53:12 nsandriushin systemd[1]: Starting dnf makecache...
May  3 21:53:13 nsandriushin dnf[44367]: Rocky Linux 9 - BaseOS                                12 kB/s | 4.1 kB
00:00
May  3 21:53:13 nsandriushin dnf[44367]: Rocky Linux 9 - AppStream                            13 kB/s | 4.5 kB
00:00
May  3 21:53:14 nsandriushin dnf[44367]: Rocky Linux 9 - Extras                            6.5 kB/s | 2.9 kB
00:00
May  3 21:53:14 nsandriushin dnf[44367]: Metadata cache created.
May  3 21:53:14 nsandriushin systemd[1]: dnf-makecache.service: Deactivated successfully.
May  3 21:53:14 nsandriushin systemd[1]: Finished dnf makecache.
May  3 21:53:14 nsandriushin systemd[1]: dnf-makecache.service: Consumed 1.192s CPU time.
May  3 21:53:23 nsandriushin gnome-shell[1844]: libinput error: client bug: timer event4 debounce short: s
cheduled expiry is in the past (-2ms), your system is too slow
[nsandriushin@nsandriushin ~]$ sudo tail /var/log/http/error_log
tail: cannot open '/var/log/http/error_log' for reading: No such file or directory
[nsandriushin@nsandriushin ~]$ sudo tail /var/log/audit/audit.log
type=USER_ACCT msg=audit(1746298481.040:355): pid=44442 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:accounting grantors=pam_unix,pam_localuser acct="nsandriushi
n" exe="/usr/bin/sudo" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="nsandriushin" AUID="nsandriu
shin"
type=USER_CMD msg=audit(1746298481.044:356): pid=44442 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0.c1023 msg='cwd="/home/nsandriushin" cmd=7461696C202F7661722F6C6F672F687474702F6
572726F725F6C6F67 exe="/usr/bin/sudo" terminal=pts/0 res=success'UID="nsandriushin" AUID="nsandriushin"
type=CRED_REFR msg=audit(1746298481.044:357): pid=44442 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:setcred grantors=pam_env,pam_fprintd acct="root" exe="/usr/b
in/sudo" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="nsandriushin" AUID="nsandriushin"
type=USER_START msg=audit(1746298481.067:358): pid=44442 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:session_open grantors=pam_keyinit,pam_limits,pam_systemd,pa
m_unix acct="root" exe="/usr/bin/sudo" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="nsandriushin"
AUID="nsandriushin"
type=USER_END msg=audit(1746298481.121:359): pid=44442 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:session_close grantors=pam_keyinit,pam_limits,pam_systemd,pa
m_unix acct="root" exe="/usr/bin/sudo" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="nsandriushin"
AUID="nsandriushin"
type=CRED_DISP msg=audit(1746298481.126:360): pid=44442 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:setcred grantors=pam_env,pam_fprintd acct="root" exe="/usr/b
in/sudo" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="nsandriushin" AUID="nsandriushin"
type=USER_ACCT msg=audit(1746298496.429:361): pid=44451 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
ned_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:accounting grantors=pam_unix,pam_localuser acct="nsandriushi
n" exe="/usr/bin/sudo" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="nsandriushin" AUID="nsandriu
shin"
type=USER_CMD msg=audit(1746298496.430:362): pid=44451 uid=1000 auid=1000 ses=3 subj=unconfined_u:unconfi
```

Рис. 2.9: Перезагрузка службы

Для того, чтобы получилось зайти, достаточно добавить 81 порт в semanage. После Этого вернём всё как было (рис. 2.10).

```

[nsandriushin@nsandriushin ~]$ sudo semanage port -a -t http_port_t -p tcp 81
Port tcp/81 already defined, modifying instead
[nsandriushin@nsandriushin ~]$ semanage port -l | grep http_port_t
ValueError: SELinux policy is not managed or store cannot be accessed.
[nsandriushin@nsandriushin ~]$ sudo semanage port -l | grep http_port_t
http_port_t          tcp      81, 80, 81, 443, 488, 8008, 8009, 8443, 9000
pegasus_http_port_t  tcp      5988
[nsandriushin@nsandriushin ~]$ service httpd restart
Redirecting to /bin/systemctl restart httpd.service
[nsandriushin@nsandriushin ~]$ sudo chcon -t httpd_sys_content_t /var/www/html/test.html
chcon: failed to change context of '/var/www/html/test.html' to 'unconfined_u:object_r:httpd_sys_content_t:s0': Operation not permitted
[nsandriushin@nsandriushin ~]$ sudo chcon -t httpd_sys_content_t /var/www/html/test.html
[nsandriushin@nsandriushin ~]$ sudo nano /etc/httpd/conf/httpd.conf
[nsandriushin@nsandriushin ~]$ sudo semanage port -d -t http_port_t -p tcp 81
[nsandriushin@nsandriushin ~]$ sudo rm /var/www/html/test.html
[nsandriushin@nsandriushin ~]$

```

Рис. 2.10: Добавление порта и возврат к первоначальной конфигурации

При добавленном порте сайт работает, но нам нужно указать, что мы обращаемся именно к 81 порту (рис. 2.11).

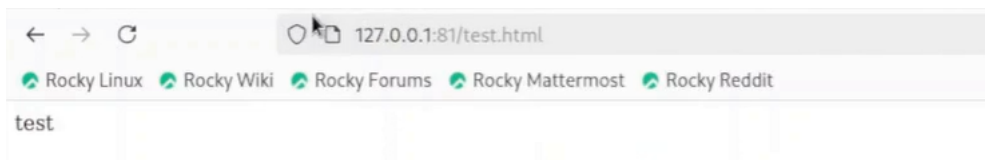


Рис. 2.11: Работающий сайт

Меняем теперь порт обратно (рис. 2.12).

```

GNU nano 5.6.1 /etc/httpd/conf/httpd.conf Modified
#
# This is the main Apache HTTP server configuration file. It contains the
# configuration directives that give the server its instructions.
# See <URL:http://httpd.apache.org/docs/2.4/> for detailed information.
# In particular, see
# <URL:http://httpd.apache.org/docs/2.4/mod/directives.html>
# for a discussion of each configuration directive.
#
# See the httpd.conf(5) man page for more information on this configuration,
# and httpd.service(8) on using and configuring the httpd service.
#
# Do NOT simply read the instructions in here without understanding
# what they do. They're here only as hints or reminders. If you are unsure
# consult the online docs. You have been warned.
#
# Configuration and logfile names: If the filenames you specify for many
# of the server's control files begin with "/" (or "drive:/" for Win32), the
# server will use that explicit path. If the filenames do *not* begin
# with "/", the value of ServerRoot is prepended -- so 'log/access_log'
# with ServerRoot set to '/www' will be interpreted by the
# server as '/www/log/access_log', whereas '/log/access_log' will be
# interpreted as '/log/access_log'.
#
#
# ServerRoot: The top of the directory tree under which the server's
# configuration, error, and log files are kept.
#
# Do not add a slash at the end of the directory path. If you point
# ServerRoot at a non-local disk, be sure to specify a local disk on the
# Mutex directive, if file-based mutexes are used. If you wish to share the
# same ServerRoot for multiple httpd daemons, you will need to change at
# least PidFile.
#
ServerRoot "/etc/httpd"
#
#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts. See the httpd.service(8) man
# page for more information.
#
Listen 12.34.56.78:80
Listen 80

```

Рис. 2.12: Смена порта обратно

3 Выводы

В результате выполнения лабораторной работы было получено понимание работы мандатного разграничения доступа

Список литературы